Testing functional immune protection in edited human beta cells

An evidence informed Perspective on endpoints and experimental design

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Abstract

Can an edit preserve useful glucose-responsive secretion in a human beta-cell preparation under immune injury? This Perspective presents an evidence-informed decision framework for testing functional immune protection. Its primary product-level outcome is sufficient regulated output per initially assigned beta-cell dose. Increased cell retention and preserved secretion per survivor are separate claims: either can explain a functional benefit, and extra retention is not a universal prerequisite. RNLS, HIVEP2 and related nominees lack a qualifying paired post-immune basal/stimulated secretion comparison in the assessed material. Other engineered human islet products provide bounded evidence connecting immune modification and function, including preclinical graft studies and one donor-islet recipient with function reported through 14 months without immunosuppression. Those findings do not establish efficacy for the nominees. An accounting identity links total glucose-induced output to retained live-cell abundance and output per survivor. Hypothetical examples show both improved retention with less output and preserved secretion without extra retention. The framework addresses misleading shortcuts: combining different preparations, relying on stimulation ratios, discarding complete losses, and inferring within-preparation success from group averages. It specifies paired endpoints, overlapping descriptive dimensions, claim-specific preservation tests and a practical calibration pathway. RNLS provides an informative case study rather than an established therapeutic lead. The framework separates comparative benefit, adequate regulated output and reliable performance within preparations over time; acceptance criteria require prospective calibration.

Introduction

A replacement beta-cell product must deliver regulated insulin at a useful dose while resisting the immune conditions into which it is placed. More persistent cells are valuable only to the extent that they retain sufficient function, or compensate for reduced per-cell performance through enough functioning mass. Conversely, a small responsive survivor population can look satisfactory after normalization while supplying little total insulin. These are different biological outcomes and require paired measurements.

The candidate evidence creates a tractable case for this distinction. RNLS was validated in autoimmune mouse experiments but did not protect in the mouse allograft setting studied by Cai et al. [1]. Subsequent RNLS/HIVEP2 spheroid graft studies did not establish durable protection [3]. Bode et al. nevertheless observed favorable survival-related outcomes in human RNLS-edited cultures after 72 hours of exposure to stimulated, HLA-mismatched immune cells [4]. These findings motivate a question about the duration, route and functional consequences of protection rather than a single success-or-failure label.

A nominated-set analysis in an allogeneic screen did not detect enrichment [2; Supplement S1], but screen rank is not a therapeutic efficacy estimate. The framework distinguishes three decisions: whether an edit outperforms an injured control, whether the product supplies sufficient regulated output, and whether that performance occurs reliably within preparations over time. Retained abundance and survivor function help interpret these decisions, independently of any particular screen result.

The broad possibility is no longer entirely hypothetical. Studies of other human islet products include post-challenge function and a recipient with a functioning modified donor-islet graft without immunosuppression [7–9]. Those results provide biological benchmarks. They do not transfer efficacy to a different edit, cell source, immune challenge or manufacturing process. Our focus is the evidence needed to move from a survival-associated perturbation to a product that sustains useful glucose-responsive output.

Framework scope

Evidence scope and interpretation

We used an experiment-level evidence map covering eleven autoimmune-screen nominees, six of which had scored human counterparts in the allogeneic dataset. The underlying alias-aware database searches, mapping rules and comparison records are retained with the screen analysis [Supplement S1]. For the present question, we added focused primary-source checks through 25 October 2026 for RNLS mechanisms and human islet products with post-immune or post-transplant function. This targeted extension is not a systematic survey of every immune-engineering strategy. The source ledger records study identity, cell product, challenge, duration, endpoint and access level; successive reports from the same recipient are treated as one clinical trajectory.

A qualifying nominee comparison requires candidate-specific edited and matched control preparations, a relevant immune challenge, post-challenge basal and stimulated secretion, and enough information to assess control responsiveness and absolute output. A report can be informative without satisfying every requirement. We distinguish cell persistence, glucose responsiveness, a controlled protection effect and adequate therapeutic output. We do not equate human cells studied in a mouse with a human clinical experiment. We did not pool heterogeneous models, infer individual measurements from plotted bars, or convert nondetection into equivalence.

Bode 2023 was assessed using its complete main text, main figure legends and all six publisher supplementary files; no direct secretion assay was located in those assessed materials [6]. The 2025 and 2026 clinical reports were accessible as primary journal abstracts and bibliographic records; their complete texts and supplements were not obtained [7,8]. We use the reported durations and broad functional findings without extracting unverified numerical trajectories. No open individual clinical or paired functional data table suitable for a new efficacy estimate was identified in the focused checks. The new numerical results below are exact analytical examples, explicitly separate from observations.

Endpoint quantities

Consider one independently prepared, assigned beta-cell product aliquot or graft. Let $N0$ denote the viable beta-cell dose at the start of immune exposure, $N(t)$ the viable beta-cell abundance at assessment, and $QL(t)$ and $QH(t)$ the absolute low- and high-glucose secretion rates measured over matched assay intervals. Rates must be attributable to living beta cells, with medium background and passive release from dying cells addressed. Define the glucose-induced increment $D(t) = QH(t) - QL(t)$, retained live-cell abundance $S(t)$, output per survivor $f(t)$, and output per initial dose $F(t)$.

$$D = QH - QL; \quad S = N / N0; \quad f = D / N; \quad F = D / N0 = S \times f. \qquad (1)$$

Equation 1 is an accounting identity, not a model of causal mediation. It applies within a preparation and requires $N > 0$ for f. If a preparation has no living beta cells, its viable-cell output is zero and f is undefined; that failure is retained in the total-output analysis. A negative or unresponsive glucose increment is reported as such, not silently truncated to a favorable value.

N/N0 is not a pure survival probability when proliferation or identity changes occur. Cell number must also be stable during the secretion assay; otherwise phase-specific live-cell time and the corresponding rates are required.

These quantities are not equally observable in every setting. In culture or perifusion, collected hormone amount over a known interval, with measured volume or flow, can estimate secretion rates; calibrated live beta-cell counts supply the denominator. Circulating insulin or C-peptide concentrations in graft studies are functional readouts, not QL or QH themselves: distribution, clearance and source attribution require additional estimation assumptions. Clinical graft abundance is usually not directly countable, and imaging or biopsy is not automatically a complete live-cell census. Such studies should report the measured proxies and their limitations rather than claim direct measurement of F or f from concentration alone.

Evidence informing the framework

The nominated edits do not yet resolve the joint endpoint

The assessed nominee literature contains baseline secretion observations for RNLS and HIVEP2 but no qualifying candidate-specific post-immune basal/stimulated comparison. The evidence accounting is supplied in Supplement S1. Bode 2023 reports metabolic, immune and graft outcomes without a located direct secretion assay in the assessed main text, figure legends and supplements [6]. This is an evidentiary gap, not a biological failure rate.

RNLS has distinct forms of supportive evidence. Cai's human stem-cell-derived beta-cell preparations responded to glucose at baseline, with no impairment reported for RNLS loss [1]. Bode's human co-culture experiment found more live CD45-negative cells after short immune exposure [4]. The latter endpoint is not beta-cell-marker-gated and does not show what the survivors secreted. Karaoglu's negative graft results constrain claims of durable protection, while their human spheroid baseline secretion cannot supply the missing post-graft measurement [3]. The findings remain compatible with early protection, delayed rejection or functionally inadequate persistence.

Mechanistic evidence sharpens the alternatives. Genetic RNLS disruption shifted metabolism toward glycolysis in mouse and human beta-cell systems, whereas pharmacological inhibition did not strongly reproduce the genetic phenotype [5]. The human planar cultures in that study lacked a glucose response in both control and edited conditions, so they cannot establish an RNLS-specific secretion defect. Bode's observations of altered immune-cell activity and candidate immunoregulatory signals support an immune-interaction hypothesis, with mechanistic necessity established only for particular tested settings [4]. Stress adaptation and immune modulation could preserve function, impose a functional cost, or have overlapping effects with another intervention. None of these possibilities is established by combining the separate experiments.

Other human islet products provide bounded evidence of coexistence

Gerace et al. provide a direct preclinical bridge. Human B2M-deficient stem-cell-derived islets transplanted into diabetic immunodeficient mice retained residual glucose-responsive insulin secretion seven weeks after mismatched human PBMC administration, whereas control grafts lost function. The GSIS assessments included five animals per group, with plasma insulin measured before and 30 minutes after glucose [9]. Protection was incomplete and rejection was delayed. This is a human-cell, mouse-host experiment connecting immune exposure and function in one graft model. Individual pre/post pairing was not verified from an animal-level table. It does not establish long-term immune tolerance or efficacy of RNLS.

Carlsson et al. reported gene-modified primary donor islets in one person with type 1 diabetes who received no immunosuppression. The product combined B2M and CIITA disruption with CD47 expression; glucose-responsive C-peptide was reported during the initial 12-week observation [7]. The subsequent journal abstract reports beta-cell function and no detected immune response against the allograft through 14 months [8]. These are successive observations in one recipient, not replicated trials. They support persistence with secretion in a human immune environment. They do not establish insulin independence, an adequate replacement dose, a causal effect relative to a randomized unedited graft, or broad safety and durability across recipients.

The benchmark evidence also includes primary human hypoimmune pseudoislets and human iPSC-derived organoids. Hu et al. reported survival, endocrine function and diabetes amelioration in allogeneic humanized mice, with avoidance of autoimmune killing reported in a separate autologous humanized model; only the abstract was assessed here [14]. Yoshihara et al. reported glycemic control beyond 50 days with PD-L1-expressing organoids in immunocompetent mouse xenografts, and glycemic and human C-peptide benefits in humanized mice [15]. These studies support context-specific connections between immune modification and function, without directly identifying secretion per surviving beta cell or establishing efficacy for RNLS.

Table 1. Selected evidence connecting persistence and function. The final two rows concern interventions outside the eleven nominees; further primary-islet and organoid benchmarks are described in the text [14,15]. Duration, cell source and evidentiary strength are not interchangeable.

Evidence setting	Observed connection	Boundary
RNLS human co-culture [4]	Favorable live CD45-negative counts after 72 h	No beta-specific paired post-immune secretion
RNLS/HIVEP2 spheroids [3]	Baseline responsiveness in human spheroids; graft protection not durable	Baseline and graft outcomes do not establish joint retention
B2M-deficient human SC islets [9]	Residual glucose-responsive insulin at 7 weeks after PBMC challenge	Mouse host; incomplete protection and delayed rejection
Modified donor human islets [7,8]	Glucose-responsive C-peptide at 12 weeks; function reported through 14 months	One recipient; later report abstract-level here; useful-dose efficacy unestablished

Endpoint framework and illustrative examples

The missing measurement leaves a sharp identification gap

Suppose a study shows retained live beta-cell abundance after immune exposure and positive secretion before exposure. Even granting nonnegative post-immune glucose-induced output, those observations allow fpost = 0. Equation 1 therefore admits Fpost = 0 despite S > 0 and fbaseline > 0. Zero is the sharp lower bound under those assumptions: a fully unresponsive survivor population is compatible with the observed quantities. The result does not predict that population; it shows why the measurements cannot rule it out.

A positive lower bound requires additional evidence or a justified assumption, such as fpost ≥ rho × fbaseline for a specified rho > 0 in the same preparation. That yields Fpost ≥ S × rho × fbaseline. Baseline measurements alone do not validate rho. Likewise, favorable survival in one batch and favorable secretion in a different batch cannot identify their coexistence without a transportability assumption. Measuring both in the challenged product removes that particular inferential gap.

How much functional loss can a survival gain conceal

For positive, comparable glucose increments and a common initial dose, let RN be the edited-to-control retained-abundance ratio and Rf the corresponding ratio of incremental secretion per survivor. The total functional ratio is RF = RN × Rf. Greater output requires Rf > 1/RN. An edit that doubles retained abundance can therefore produce less glucose-induced output if its per-survivor increment falls below one-half of the control value. The boundary is algebraic; it is not a clinical acceptance threshold.

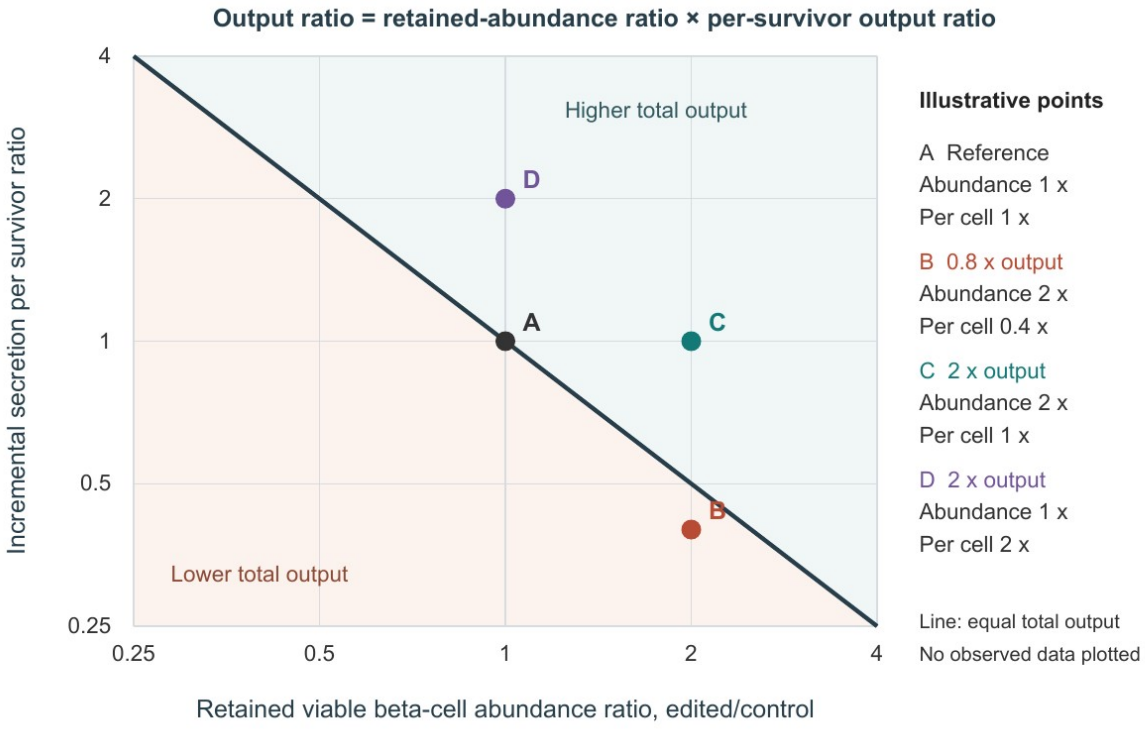


Figure 1. Total output and its descriptive components. The line RN × Rf = 1 separates higher from lower glucose-induced output at equal initial dose; ratios use each example's matched challenged control. A denotes a control reference. B doubles retained abundance but produces 0.8 times the output; C doubles abundance with preserved per-survivor increment. D doubles output through a larger per-survivor increment with unchanged retained abundance. These relative positions do not establish adequate output, acceptable basal release, causal mechanism or durability. All points are hypothetical.

The illustrative preparations begin with 100 beta cells. A challenged control retains 40 cells, each contributing basal and stimulated rates of 0.5 and 1.5 arbitrary units: total rates are 20 and 60, with an increment of 40. An edited preparation retains 80 cells, each contributing 0.2 and 0.6 units: total rates are 16 and 48, with an increment of 32. Both have a stimulation index of 3, yet the preparation with twice as many living cells supplies 20% less glucose-induced output. Molano et al. have already demonstrated the practical importance of distinguishing absolute insulin output from stimulation index in human islet potency assessment [10].

A separate hypothetical comparison illustrates protection without extra retention. Both preparations start with 100 cells and a per-cell increment of 1 unit; both matched unchallenged preparations retain these values at assessment. After immune challenge, both retain 40 cells, but the control increment falls to 0.5 units per survivor while the edited increment remains 1. Total increments are 20 and 40, and F is 0.2 versus 0.4, with no unchallenged edit advantage. The edited preparation supplies twice the output despite identical retained abundance (Figure 1D). This is

compatible with prevention of secretory dysfunction; it does not prove the cellular mechanism or establish an adequate product dose.

Paired measurements remain necessary at the group level

Averaging does not remove the problem. In general, mean(S × f) is not mean(S) × mean(f); the association between retained abundance and survivor output matters. For preparations with a common initial dose, an exact aggregate decomposition uses abundance-weighted survivor output, fweighted = sum(D)/sum(N), alongside mean retained abundance. The primary functional comparison should therefore be estimated directly from each assigned preparation's F, retaining complete-loss outcomes. The observed function of survivors is a useful product-quality description, but conditioning on post-treatment survival does not by itself identify the edit's causal effect on the same cells [13].

Even favorable group means can hide nonoverlap: one batch may retain many poorly responsive cells while another retains a few highly responsive cells. Evaluate a joint product-success indicator within each preparation, using prespecified adequate output, acceptable basal release, regulated response and persistence criteria, and report its frequency across independent preparations. Additional abundance or per-survivor preservation criteria belong to explicitly named claims, not automatically to every product-success definition. This prevents coexistence from being assembled out of different preparations while allowing function to be protected without an abundance gain.

A discriminating test of functional protection

One product and a blocked challenge comparison

RNLS is an informative first test because it combines favorable short-term human culture survival, responsive baseline preparations and adverse graft findings. Prioritizing it is an expected-information judgment, not a ranking of therapeutic efficacy. The primary comparison should use independently generated RNLS-edited and matched control derivatives of the same glucose-responsive human beta-cell product, with edit verification and a rescue or independent perturbation to distinguish on-target biology from preparation effects. Starting viable beta-cell dose, maturation and handling should be matched; manufacturing yield and baseline failures should also be reported.

Within each independent target-cell source and preparation batch, allocate matched aliquots to unchallenged, autoimmune and allogeneic conditions. Autoimmune exposure requires verified antigen-specific recognition in an appropriate HLA context; generic inflammatory stress is not an autoimmune challenge. Allogeneic exposure requires documented target-effector mismatch. Qualify each challenge by demonstrable functional injury or cell loss in controls, without immediately destroying every group; secretory injury alone can be a relevant endpoint. Use the same target product and measurement workflow across arms, but estimate each immune contrast separately. The effector populations and injury mechanisms cannot be made identical simply by matching assay duration.

Assess baseline, early injury, repeated or sustained exposure, and prespecified later intervals. The final duration must be chosen for the model and intended use before results are inspected; a 72-hour culture benefit cannot be relabeled durable graft protection. Pair secretion with beta-cell-specific live abundance in the same challenged aliquot or graft whenever possible. If mass measurement is destructive, prespecify terminal paired assays or clearly identified matched aliquots; do not describe separate batches or sacrificed specimens as longitudinal measurements of the same cells. Repeated glucose testing must itself be checked for perturbation.

Measure output and survivor state separately

At each functional assessment, report absolute basal output, stimulated output, their difference and the time course of return toward basal secretion. Measure the viable beta-cell denominator with markers capable of detecting identity shifts; CD45 exclusion, total DNA, graft volume or total insulin content alone are insufficient substitutes. Confirm that the increment reflects regulated release rather than injury-related leakage. Dynamic secretion and beta-cell identity measurements distinguish a responsive survivor population from altered cellular composition. If an edit changes cell cycling, report proliferation alongside retained abundance.

For RNLS, metabolic and identity measurements can test whether a putative survival advantage accompanies an altered functional state. They should be obtained in paired or explicitly matched samples before and after challenge. A glycolytic signature is not itself proof of dedifferentiation, impaired stimulus–secretion coupling or protection. Functional rescue, independent perturbations and time-resolved observations would be required to support a mechanistic attribution. A candidate that merely postpones the same decline would show an early abundance advantage followed by loss of output during continued exposure.

A joint decision rule with calibrated margins

The primary biological estimand is the edited-versus-control difference in glucose-induced output per initial dose under each immune challenge at a prespecified late assessment. Estimate absolute stimulated output and retained live abundance alongside it. An edit-by-challenge contrast, comparing that difference with its unchallenged counterpart, helps separate challenge-related protection from a baseline growth or secretion advantage. Neither interaction alone nor a favorable comparison against an almost nonfunctional control establishes an adequate product. Compare output with both matched edited/unchallenged and unedited/unchallenged preparations, and with a separately specified responsive reference and minimum useful output. Prespecify an acceptable basal-output range or ceiling as well: a positive increment does not rule out excessive unregulated secretion.

Table 2. Claim-specific decision requirements. Descriptive dimensions may overlap. Biological acceptance criteria, reference product, duration and multiplicity plan must be fixed before confirmatory testing.

Claim	Required comparison	Interpretation
Product-level functional protection	Regulated output and F under challenge versus matched control, both unchallenged genotypes and a responsive reference	Require sufficient output, acceptable basal release and challenge-related benefit over the declared period
Increased cell retention	Challenged edited versus control N/N0, with the unchallenged contrast	Separate abundance claim; examine growth and identity changes; no universal requirement for an abundance gain
Preserved survivor function	Paired f and dynamic response against a responsive reference with a justified margin	Failure to demonstrate per-survivor noninferiority does not automatically negate adequate product-level output.
Within-preparation product success	Adequate regulated output and persistence in the same preparation; add criteria only for the stated claim	Report joint success and uncertainty; favorable marginal means do not establish coexistence

Product-level functional immune protection means preserving sufficient regulated output under immune challenge, with benefit supported by the matched challenge comparison over the stated

period. It does not universally require more retained cells or per-survivor noninferiority. Increased retention, improved or preserved survivor output, and adequate total product output are separate claims. Their observed components describe how the output difference arises; a causal mechanism requires additional evidence.

These are overlapping dimensions, not mutually exclusive categories. A larger population with a modestly lower per-survivor increment may have compensated output and also satisfy a justified noninferiority margin. Conversely, a product can deliver sufficient regulated output while failing a stricter survivor-preservation claim. Report each claim separately. The narrower label survival-mediated functional protection requires a retention advantage as well as product benefit, with mechanistic attribution justified separately. Persistence without useful function and early benefit followed by late failure remain informative outcomes.

Measurement and calibration before confirmatory testing

First specify the intended product use, starting dose and observation period. These determine what minimum output and duration would be biologically meaningful; a clinical replacement objective ultimately requires dose-appropriate glucose control and patient outcomes. Select an independent, glucose-responsive reference preparation and use its validated basal/stimulated profile and response kinetics to anchor assay acceptance and a basal-output range. Include both matched unchallenged genotypes so an intrinsically weak edited baseline cannot make retention appear adequate.

Next define the largest biologically acceptable loss for each preservation claim using external functional or dose-response evidence and the intended use. Assay precision determines whether that loss can be measured reliably; it does not determine how much loss is acceptable. Characterize repeatability, quantification limits and recovery, then use pilot independent preparations and immune donors to estimate between-preparation variance, covariance, attrition and challenge stability. Those estimates inform sample size, measurement frequency and power for the declared joint claim. An imprecise assay should be improved, rather than used to justify a more permissive biological margin.

Finally justify the required proportion of preparations meeting product criteria from intended manufacturing reliability and use, and select late assessments from model kinetics and the intended duration of protection. Lock the reference, output and basal limits, any claim-specific noninferiority margin, duration, joint-success benchmark and analysis plan before confirmatory candidate results are inspected. A nonsignificant difference does not establish preservation [11,12]. The present specification leaves these values unresolved and is therefore an experimental framework, not an executable confirmatory protocol. In vitro potency criteria do not by themselves establish clinical utility.

Independent cell sources, preparation batches and immune donors supply biological replication. Technical wells and repeated measurements do not increase the number of independent products. Use blocked allocation and an analysis reflecting crossed target and immune donors, nested preparations, and repeated time points. Estimate output uncertainty from paired preparation-level measurements. Power the prespecified joint claim using pilot between-preparation variance, covariance, expected attrition and a meaningful effect; the current evidence does not justify a numerical sample size.

For one fixed claim requiring every component to pass, an intersection–union rule can test each component at the same one-sided alpha and declare the conjunction only when all pass. Marginal mean-endpoint tests alone cannot establish within-preparation coexistence. That claim also requires prespecified inference for the joint-success frequency against its own justified benchmark, with uncertainty and an edited-control comparison when claiming benefit. The conjunction rule

does not create a simultaneous confidence region or permit unadjusted selection across multiple edits, time points or challenge settings. Report component estimates and a multiplicity plan for separate or selected claims. Complete loss is an output failure, while unavailable measurements require explicit missing-data sensitivity analysis. A high per-survivor estimate must never conceal dropped dead cultures.

Discussion

The evidence supports a conditional answer. Edited human islet products can retain function after immune exposure, and one modified donor-islet graft has been reported to function without immunosuppression beyond a year [7–9]. This supplies proof-of-concept support for coexistence. It does not show that the original nominees, or any particular RNLS implementation, produce durable and therapeutically adequate functional protection. It also does not turn an uncontrolled recipient trajectory into a comparative efficacy estimate.

For RNLS, candidate advancement depends on separating relative benefit, adequate output and reliable performance over time. An edit can outperform a severely injured control while leaving output inadequate. Useful output can coexist with reduced per-survivor performance, and favorable averages can conceal inconsistent preparation-level success. Paired measurements and prospectively calibrated criteria determine which advancement decision the evidence supports.

A combination strategy is a subsequent question rather than a presumed solution. If a candidate retains function but provides incomplete protection, a factorial comparison of the candidate, an independently characterized immune-modifying intervention, their combination and the matched control can test complementarity. RNLS-associated immune signals may overlap with other modifications, so additivity cannot be assumed. The same product-output requirements and separately declared retention or survivor-quality claims apply to the combination.

The present study has deliberate limits. The evidence extension is focused and may miss relevant reports. Clinical experience summarized here is limited to one recipient and dependent follow-ups, assessed through primary abstracts. Post-transplant C-peptide supports graft function but does not directly quantify secretion per surviving beta cell. Model duration is not lifetime durability. The proposed experiment has not been performed. Long-term product evaluation must also address controlled growth, identity stability and the consequences of reduced immune recognition.

The immediate research objective is to establish whether an edited human beta-cell preparation sustains sufficient regulated output under immune challenge, and then determine which retention and survivor-function claims the measurements support. The same design can identify benefit through greater retained abundance, preservation of secretion without extra retention, or compensation across those dimensions. Adequate, compensated, transient or inadequate output would support different decisions about advancing the product, modifying it, combining interventions or stopping that implementation.

Data and reproducibility

The package supplies the source ledger, endpoint definitions, hypothetical examples, figure-generating code and experimental framework. Supplement S1 identifies the retained screen and nominee evidence inputs. Eight checks verify arithmetic and boundary conditions, not biological efficacy. The calculations and editable manuscript can be regenerated from the supplied sources. The rendered PDF was visually reviewed. AI assisted retrieval, analysis, writing and internal checking.

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